

When to Truncate the Archive? On the Effect of the Truncation Frequency in Multi-Objective Optimisation

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Abstract

Using an archive to store nondominated solutions found during the search of a multi-objective evolutionary algorithm (MOEA) is a useful practice. However, as nondominated solutions of a multi-objective optimisation problem can be enormous or infinitely many, it is desirable to provide the decision-maker with only a small, representative portion of all the nondominated solutions in the archive, thus entailing a truncation operation. Then, an important issue is when to truncate the archive. This can be done once a new solution generated, a batch of new solutions generated, or even keeping all nondominated solutions generated and truncate it later. Intuitively, the last approach may lead to a better result since we have all the information in hand before performing the truncation. In this paper, we study this issue and investigate the effect of the timing of truncating the archive by using the criteria commonly used in MOEAs' population maintenance procedure. We show that, interestingly, truncating the archive once a new solution generated tends to be the best, whereas considering an unbounded archive is often the worst. Our results highlight the importance of developing effective subset selection techniques (rather than employing the population maintenance methods in MOEAs) when using a large archive¹.

CCS Concepts

• **Mathematics of computing** → **Combinatorics**; • **Theory of computation** → **Evolutionary algorithms**; • **Applied computing** → **Multi-criterion optimization and decision-making**.

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Keywords

Multi-objective optimisation, evolutionary algorithms, archiving, subset selection

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1 Introduction

Recently, there has been an increasing interest in using a large (or even unbounded) archive to store the best solutions discovered throughout the search process of an MOEA. This can be attributed to 1) advancements in the computational and memory capabilities of modern computers, and 2) the inability of the final population of MOEAs to fully represent the best solutions discovered during the search process [1–4]. This practice belongs to a broader topic, termed archiving [2, 5], which studies the process of comparing new solutions with existing solutions in the archive and determining which ones to keep or to discard.

There is an important issue in archiving – when to maintain or truncate the archive [6, 7]. Conventionally, the archive truncation can be performed either once a new solution arrives or a batch of new solutions arrive. The former corresponds to the case of updating the population once a new solution is generated (i.e., the $\mu + 1$ evolution mode) such as in SMS-EMOA [8] and MOEA/D [9], while the latter corresponds to the case of updating the population once a number of solutions are generated (i.e., the $\mu + \mu$ evolution mode) such as in NSGA-II [10], IBEA [11] and NSGA-III [12]. In addition, recently people tend to use an unbounded archive to store all non-dominated solutions found during the search [13–18]. Afterwards, the archive is truncated down to a small, but representative subset provided to the decision maker (as a large number of solutions can easily overwhelm them). Intuitively, one may think this approach is better since we have all the information in hand before performing the truncation [3]. The $\mu + 1$ approach that truncates the archive once every new solution arrives may be less promising due to the unknown about future input when the decision is made.

Table 1: IGD values (mean $\pm$ standard deviation) of the truncation methods from NSGA-II, SMS-EMOA, IBEA, MOEA/D and NSGA-III with the three truncation approaches. Statistical significance is measured by the Wilcoxon rank-sum test ($p < 0.05$), where the letter in the alphabetical order (a, b, c) indicates statistically significant superiority of one algorithm over another. That is, the approaches sharing the same letter have no statistical difference; e.g., an algorithm labelled as (a) is statistically better than an algorithm labelled as (b), and neither of them is statistically different from an algorithm labelled as (ab).

Sequence	Truncation approach	NSGA-II	SMS-EMOA	IBEA	MOEA/D	NSGA-III
Simplex	<i>immediate</i>	4.650e-02 $\pm$ 1.4e-03 (<i>a</i>)	3.798e-02 $\pm$ 1.4e-04 (<i>a</i>)	4.153e-02 $\pm$ 4.4e-04 (<i>a</i>)	3.718e-02 $\pm$ 0.0e+00 (<i>a</i>)	3.732e-02 $\pm$ 1.1e-05 (<i>ab</i>)
	<i>batch</i>	4.835e-02 $\pm$ 1.9e-03 (<i>b</i>)	3.895e-02 $\pm$ 1.8e-04 (<i>b</i>)	4.150e-02 $\pm$ 5.3e-04 (<i>a</i>)	3.718e-02 $\pm$ 0.0e+00 (<i>a</i>)	3.732e-02 $\pm$ 7.6e-06 (<i>a</i>)
	<i>unbounded</i>	2.229e-01 $\pm$ 0.0e+00 (<i>c</i>)	3.911e-02 $\pm$ 9.7e-05 (<i>b</i>)	4.316e-02 $\pm$ 0.0e+00 (<i>b</i>)	3.718e-02 $\pm$ 0.0e+00 (<i>a</i>)	3.732e-02 $\pm$ 0.0e+00 (<i>b</i>)
Inverted simplex	<i>immediate</i>	4.638e-02 $\pm$ 1.1e-03 (<i>a</i>)	3.806e-02 $\pm$ 1.2e-04 (<i>a</i>)	4.673e-02 $\pm$ 5.4e-04 (<i>b</i>)	6.214e-02 $\pm$ 0.0e+00 (<i>a</i>)	4.856e-02 $\pm$ 8.3e-04 (<i>a</i>)
	<i>batch</i>	4.824e-02 $\pm$ 9.5e-04 (<i>b</i>)	3.904e-02 $\pm$ 1.9e-04 (<i>b</i>)	4.641e-02 $\pm$ 6.1e-04 (<i>ab</i>)	6.214e-02 $\pm$ 0.0e+00 (<i>a</i>)	4.938e-02 $\pm$ 9.8e-04 (<i>b</i>)
	<i>unbounded</i>	2.370e-01 $\pm$ 0.0e+00 (<i>c</i>)	3.959e-02 $\pm$ 6.0e-05 (<i>c</i>)	4.633e-02 $\pm$ 0.0e+00 (<i>a</i>)	6.214e-02 $\pm$ 0.0e+00 (<i>a</i>)	5.042e-02 $\pm$ 9.3e-04 (<i>c</i>)

In this paper, we study this issue to investigate the effect of the timing/frequency to perform the truncation to the archive. We consider the three approaches stated above, i.e., 1) truncating the archive once a new solution arrives, 2) truncating the archive once a batch of new solutions arrive, and 3) truncating the archive after all solutions arrive. We apply well-established criteria that are commonly used in the population update procedure of five MOEAs (NSGA-II, IBEA, SMS-EMOA, MOEA/D and NSGA-III).

2 Experimental Design

We consider three approaches to truncate the archive: 1) *Immediate*, i.e., truncating the archive once a new solution arrives; 2) *Batch*, i.e., truncating the archive once μ solutions arrive, and 3) *Unbounded*, i.e., truncating the archive after all solutions arrive. Here, μ stands for the capacity of the archive, which resembles the $(\mu + \mu)$ evolution mode in MOEAs, i.e., selecting μ solutions from 2μ solutions. We consider well-known population truncation methods from five representative MOEAs. They are NSGA-II [10], IBEA [11], SMS-EMOA [8], MOEA/D [9] and NSGA-III [12].

We consider two types of sequences, each with 5,000 solutions, sampled randomly from a simplex Pareto front and an inverted simplex Pareto front respectively. For each type of sequences, we randomly shuffled 31 times, generating 31 different orders, as the order of the sequences may affect the archiving results [19]. We then apply the truncation methods to these sequences under the three truncation frequencies. We use the Wilcoxon rank-sum test to measure statistical soundness and consider the run with the median quality for visualisation. We evaluate the quality of truncation using the IGD indicator [20], with a reference set of 5,050 evenly distributed points on the (inverted) simplex. The archive capacity μ is set to 105 to ideally obtain a uniformly distributed solution set.

3 Results

3.1 NSGA-II

Since all solutions in the sequences are nondominated, the quality of solutions is only determined by crowding distance of NSGA-II. As can be seen in Table 1, the *immediate* truncation approach obtains the best IGD value, followed by the *batch* approach. The *unbounded* approach has the worst result, falling far behind the other two (nearly an order of magnitude). This can be confirmed in Figure 1, where solutions under the three truncation frequencies in a run with the median IGD value are plotted.

As we can see in the figure, the *unbounded* truncation fails to cover the Pareto front, with their solutions clustering in the three corners. This is because NSGA-II uses a “one-off” way to perform

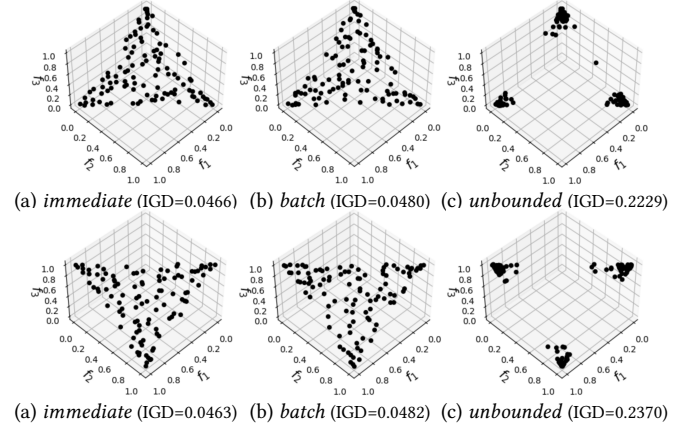


Figure 1: Solutions obtained by NSGA-II under the three truncation approaches on simplex-shaped (top panel) and the inverted simplex-shaped (bottom panel) sequences in the run with the median IGD.

the population update [10], i.e., it calculates the crowding distance of each solution at the beginning and then places μ solutions (here $\mu = 105$) with the best crowding distance into the archive (i.e., effectively removing the $(5000 - 105)$ worst solutions at one time). This practice, which does not update the crowding distance after a solution removed in the set, may lead to poor diversity (since the crowding distance is no longer accurate) [21]. In the *unbounded* truncation approach, this effect will become much more pronounced since there are 4,895 solutions to be removed.

In addition, it is worth mentioning that regardless of the truncation approaches, uniformity of the solutions obtained by NSGA-II is poor. This is because the crowding distance fails to measure the crowdedness level of solutions in the space with three or more objectives – an issue which has been frequently reported and studied in the area, theoretically [22] and empirically [23, 24].

3.2 SMS-EMOA

The results obtained by the truncation method of SMS-EMOA are given in Table 1 and Figure 2. As we can see, the three truncation approaches can well maintain the solutions’ uniformity, with their solutions being distributed fairly evenly on the Pareto front.

However, interestingly, the *immediate* approach performs best, followed by the *batch* approach, and *unbounded* approach performs the worst (as can be seen from their IGD values). This sounds a bit counter-intuitive since the *unbounded* approach has all the solutions available, while the *immediate* one can only see one-step ahead (i.e., only one solution available). One possible explanation is

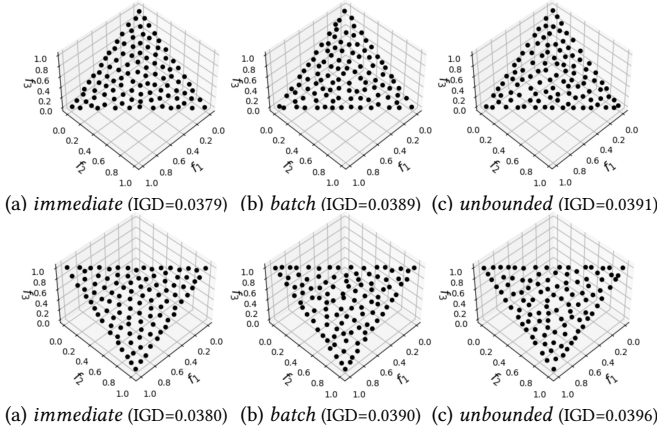


Figure 2: Solutions obtained by SMS-EMOA under the three truncation approaches on simplex-shaped (top panel) and inverted simplex-shaped (bottom panel) sequences in the run with the median IGD.

that the truncation method in SMS-EMOA still uses the one-by-one greedy truncation, removing the solution contributing least to the set's hypervolume. Such a greedy truncation may introduce error (when the number of solutions to be truncated is larger than 1). In SMS-EMOA, for the *unbounded* approach there are (5000 - 105) iterations, i.e., 4895 solutions needed to be removed. Consequently, the error will be accumulated and become pronounced after many iterations. This phenomenon is actually known as *nesting effect* in the area of subset selection [25]. In contrast, the accumulated error should be smaller in the *batch* approach, and even it is none in the *immediate* approach.

3.3 IBEA

In IBEA, similar results are obtained under the three truncation approaches for both simplex and inverted simplex sequences (Figure 3). It can be seen that the solutions are not distributed as uniformly as in SMS-EMOA, more crowded along the boundary lines of the Pareto front, particularly for the inverted simplex sequences.

It is worth mentioning that the nesting effect observed in SMS-EMOA also applies to IBEA – the one-by-one greedy truncation may accumulate larger error with more iterations. However, this seems to be largely overshadowed by the algorithm's behaviour of boundary solution preference. This may explain why unlike SMS-EMOA's clear performance degradation pattern over the increase of the archive size to be truncated from, IBEA shows less pronounced archive-size-dependent behaviour.

3.4 MOEA/D

The results of MOEA/D under the three truncation frequencies are given in Table 1 and Figure 4. MOEA/D turns out to be the algorithm invariable to the truncation frequencies. As can be seen in the table, it has the same IGD means and zero standard deviation, meaning that the algorithm acquires the same solutions over all the 31 sequences under the three truncation approaches. Regarding the solution quality on the sequences with different shapes, it is well known that MOEA/D performs perfectly on a simplex Pareto front (as shown in the top panel of Figure 4), whereas it performs poorly on a Pareto front with an irregular shape like the inverted simplex (the bottom panel of Figure 4) [26].

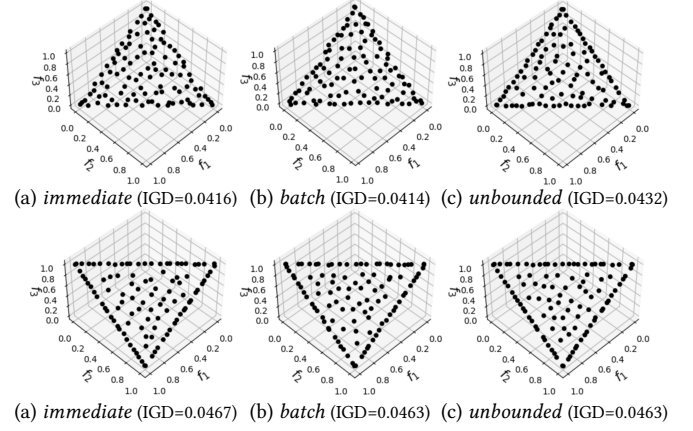


Figure 3: Solutions obtained by IBEA under the three truncation approaches on the simplex-shaped (top panel) and inverted simplex-shaped (bottom panel) sequences in the run with the median IGD.

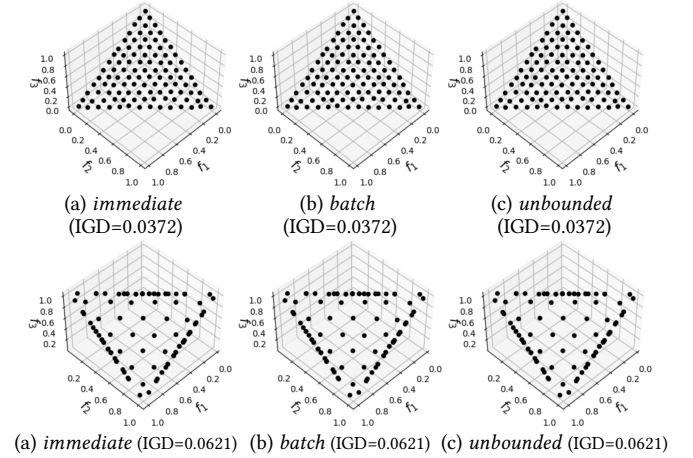


Figure 4: Solutions obtained by MOEA/D under the three truncation approaches on the simplex-shaped (top panel) and inverted simplex-shaped (bottom panel) sequences in the run with the median IGD.

3.5 NSGA-III

The results of NSGA-III under the three truncation frequencies are given in Table 1 and Figure 5. As can be seen, on the simplex-shape sequence, all the three truncation approaches perform very similarly, with an identical mean IGD value and the same solutions obtained by the run with median IGD. However, for inverted simplex sequences, the *immediate* performs best, followed by the *batch* approach, and the worst is the *unbounded* approach. This is unexpected since NSGA-III randomly chooses solutions associated with the weight vectors after the closest match, hence differences of the solutions under the three truncation frequencies are likely to be statistically indifferent. A possible explanation lies in NSGA-III's normalisation strategy: dynamic updates to the reference points in the *immediate* approach may help preserve diversity (as different reference points correspond to different solutions) before the global ideal and nadir points are found.

4 Conclusion

This paper investigated the effect of the timing of truncating the archive: 1) once a new solution arrives (*immediate*), 2) once a batch

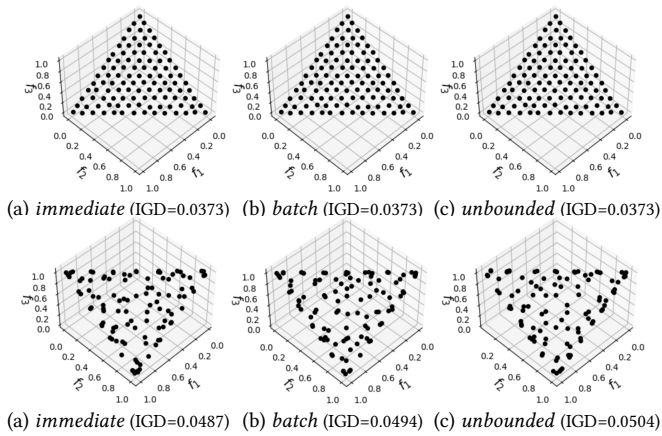


Figure 5: Solutions obtained by NSGA-III under the three truncation approaches on the simplex-shaped (top panel) and inverted simplex-shaped (bottom panel) sequences in the run with the median IGD.

of new solutions arrive (*batch*), and 3) after all solutions arrive (*unbounded*). We considered the truncation methods from five representative MOEAs. Our findings are summarised as follows.

- The one-off truncation method which does not update the solution information (as in NSGA-II) is not suitable for truncating a large archive (e.g., in the *unbounded* approach).
- The one-by-one greedy removal method which removes the worst solutions iteratively (used in SMS-EMOA and IBEA) is not suitable for the *unbounded* approach.
- Truncating the archive by removal tends to perform slightly better than by inclusion (i.e., adding the best solutions iteratively into the archive) when the number of solutions to be truncated down to is commensurate with the total solution number (e.g., in the *batch* approach).
- For many MOEAs (NSGA-II, SMS-EMOA and IBEA), the *immediate* approach is in general the best, followed by the *batch*, and the *unbounded* is the worst; while for some other MOEAs (MOEA/D and NSGA-III), the effect of the timing of truncating the archive is minor, if any.

In this study, we consider the sequences consisting of solutions randomly sampled from the Pareto front, i.e., they are nondominated to each other. Another way is to consider the sequences obtained by an MOEA, e.g., initial solutions, offspring solutions in the 1st generation, offspring solutions in the 2nd generation, etc. We expect that the second way may obtain different results as the Pareto dominance relation will play a role here for the *immediate* and *batch* approaches – the archive may deteriorate, i.e., it may accept solutions which are dominated by the solutions removed previously, as the truncation methods in the considered MOEAs are all subject to deterioration [6]. We will leave this for future study.

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